PRAVAN - Chapter 1: Introduction

Fondements du Web (Web Foundations) - full lesson explained in English

Programmation Avancée, Master 1 Informatique (Big Data), USTHB - Dr. Boubenia Mohamed - session of 21/10/2021

> Covers all 48 slides. Anything not on the slides is marked **(added)**.

Course info

- **Module:** Programmation Avancée (PRAVAN), Master 1 Informatique (Big Data), USTHB.
- **Teacher:** Dr. Boubenia Mohamed, researcher-lecturer, Laboratoire des Systèmes Informatiques, Computer Science Department.
- **Session plan:** (1) module objective, (2) web foundations, (3) module content, (4) semester plan (lectures / labs / evaluation).

1. Module objective

Why a Big Data master has a web course

- The master trains people to assist company and organization decision-makers.
- It gives a double skill set: computer science plus decision support. That is valuable in the socio-economic job market and also leads to doctoral studies and research.
- The pipeline: **different data sources** → **extraction** → **dynamic web pages (dashboards)** → **presentation and distribution** → **decision-makers.**

PRAVAN's role

It covers the **dynamic web pages** step. Students get familiar with new web technologies: **Python, multimedia web, and web services**.

Final goal: design a **dynamic website** with both **client-side and server-side** features. In short: web programming.

Prior knowledge

- **L3 Informatique** students mostly already did web programming (web basics, HTML, CSS, dynamic web).
- **L3 Mathématique** students did not, so they must review web basics, HTML and CSS.
- It is not just repetition: the course also covers new things such as JavaScript and Python.

2. Foundations of the Web

Internet ≠ Web

To understand how they relate, you have to look at how each was invented.

	Internet	Web
When	1960s	1989
Components	Modem, switch, IP, DNS	HTML, HTTP, hyperlinks

- The Internet is not "having websites". The Web is all the sites you browse.
- **The Web is just one application of the Internet**, but it is the one that got the general public and media interested in the Internet.

The Internet

Goal: make several distant machines communicate.

Origins: J.C.R. Licklider (director of ARPA's IPTO, 1962-64), Robert Taylor (IPTO director, 1966-69) and Larry Roberts (IPTO chief scientist, 1966-73). Their 1968 paper "The Computer as a Communication Device" argues we are entering an era where we interact with living information, not passively like with books and libraries, but as active participants contributing to a continuous process.

Inventions needed:

Invention	What it is
Modem	Device that sends digital data (0s and 1s) over a telephone line, which is analog.
Automatic switching	If A wants to talk to D, the network must find a wired route (A-B-D or A-C-D), intelligently, resiliently and automatically.
IP addresses	A unique identifier for every machine. Format XXX.XXX.XXX.XXX with each XXX in [0, 255]. May be followed by a **port**, e.g. 63.255.173.183:8080 (the slide says this comes back later). With 4 computers (1969) it was easy; by 1980 the network was much bigger and needed a smarter strategy.
DNS (Domain Name System)	IP addresses are fine for machines but not for humans. DNS maps a memorable name to an IP (www.google.be → 172.217.9.131; www.usthb.dz instead of 193.194.89.19). Humans use domain names, machines use numeric addresses, and typing the IP directly still works.

Evolution: today the Internet runs on a worldwide network of submarine cables (slide shows a world map plus photos of cable-laying ships and divers).

The Web

- **WWW** = "toile d'araignée" (spider's web) in French. Invented in **1989 at CERN** (European Centre for Nuclear Research) by **Sir Tim Berners-Lee**.
- **The problem:** CERN had physicists from all over the world with different document formats, software and operating systems. Tim, a software engineer there, was frustrated by this heterogeneous mass of information and decided to rethink how information is structured and accessed. He built a **distributed hypertext system** called World Wide Web.
- **First web page:** http://info.cern.ch/hypertext/WWW/TheProject.html
- The 3W project is defined as a **hypermedia information-retrieval initiative**.
- **Hypertext:** clickable text that lets you navigate from one page to another.
- **Hypermedia:** hypertext that includes not just text but also images, sound and video inside a web page.

Web page vs website

- **Web page** (slide example: blender.org): made of text elements such as a title, a paragraph and a button.
- **Website:** a set of web pages stored on a computer. Usually built around a central **home page** with links to other pages on the same server (**internal links**), and sometimes **external links** to pages hosted on another server.

Accessing a website

- 1. Open a web browser (the **web client**): Chrome, Firefox, Opera, etc.
- 2. Type the site's address, called a **URL (Uniform Resource Locator)**, e.g. www.blender.org.
- 3. Result: the site opens and you get a web page.

What happens behind the scenes

Between typing the address and seeing the page there are hidden steps. You need to know them at a minimum to build a website.

Step	What happens
1	You type the site address in the **browser**.
2	The browser **requests the web page** from the **web server**.
3	The server **responds by sending the page**, which is an **HTML document**.
4	The browser **reads the HTML and displays the page**. Press **CTRL+U** to see the HTML source. The browser has "interpreted" that code.

This exchange uses **HTTP (Hypertext Transfer Protocol)**, a client-server communication protocol: the client sends a **request**, the server sends a **response** (an HTML document).

Elements of the Web (slide 33): URL, client (browser), HTTP, web server, HTML document.

HTML and CSS

- **HTML (HyperText Markup Language):** to create a web page you must learn to write an HTML document. It **fills and structures the content**: titles, lists, paragraphs, images, etc.
- Raw HTML alone looks plain, so something else is needed to make it presentable.
- **CSS (Cascading Style Sheets):** a separate CSS document controls presentation: text color, font, image positioning. The slide shows a styles.css file with rules for h1 and h2 (font-family, font-size, font-weight, font-variant, text-align, margins).

Static vs dynamic sites

Type	How it works
Static (HTML + CSS)	Browser sends an HTTP request, server replies with HTML+CSS. Enough for presentation.
Static + JavaScript (interactive)	Response is HTML + CSS + **JavaScript**, a program executed **client-side** (in the browser). Adds client-side interactivity: capturing user actions (clicks), doing calculations, validating forms; anything a programming language can do.
Dynamic	On top of that, a program runs **server-side** (Java / PHP / JavaScript / Python). Page content is **built dynamically** and can vary with information such as the time, the user's name or a submitted form, known only when the page is requested. The server may also **use a database** (SQL requests one way, data back the other way).

3. Course content

Chapter	Topic
1	Introduction (this lesson)

Chapter	Topic
2	Static site: HTML and CSS
3	Interactivity (client-side programming): JavaScript
4	Dynamic site: Python and Flask
5	Databases (MySQL)
6	Single Page App (SPA): AJAX (Asynchronous JavaScript And XML)

4. Semester plan

Dates are from the 2021-22 schedule. The Info and Math groups have staggered labs (TP), because Math students need more time.

Week	Lecture	TP Info	TP Math
1 (17 Oct)	Ch. 1 Intro	-	-
2 (24 Oct)	Ch. 2 HTML	TP1 Intro	TP1 Intro
3 (31 Oct)	Ch. 2 CSS	TP2 HTML/CSS	Start TP2
4 (7 Nov)	Ch. 3 JS	Finish TP2	TP2 HTML/CSS
5 (14 Nov)	Ch. 4 Python	TP3 JS	Start TP3
6 (21 Nov)	Ch. 4 Flask	Finish TP3	TP3 JS
7 (28 Nov)	Ch. 5 Databases	TP4 Python+Flask	Start TP4
8 (5 Dec)	Ch. 6 Ajax	Finish TP4	TP4 Python+Flask
9 (12 Dec)	Revision (with a question mark on the slide)	TP5 Databases+Ajax	TP5 Databases+Ajax
Holidays			
10 (2 Jan)		**Evaluation**	
11 (9 Jan)			**Evaluation**

Grading

- **TP continuous assessment:** /5
- **TP grade** (project to submit, or test): /15. Together the TP mark is /20.
- **EMD (written exam):** /20
- **Credits: 4, Coefficient: 4**, counted in the overall average.

Errors in the slides themselves

- Slide 12 says each IP number must be in [0, 255], then shows **151.119.354.12**. 354 is invalid, so it is a mistake on their part.
- Slide 15 has the typo "Swtich" (Switch). Slides 34 and 36 say "Langage" instead of "Language".
- Slide 21 uses "www.blender.org" while slide 19's browser shows "blender.org". Both work; not a conflict.
- **(added)** Each IP number is one byte, so an IPv4 address is 32 bits. The slides don't say this.

What matters most for an exam

- Internet vs Web, and who invented what and when.
- The 4-step browser-server flow and the roles of HTTP, URL and HTML.
- HTML = structure, CSS = style, JS = client-side behavior, Python/Flask etc. = server-side logic, database = storage.
- Static vs dynamic: what changes when server-side code and a database enter the picture.
- The chapter map and the grading split.